

# Project Delivery & Deployment Report

Date: May 17, 2026

From Source Code to Live Hosting Infrastructure

## Executive Summary

This report documents the end-to-end development, deployment, and optimization lifecycle for the Dr. Ranjith G website (**dranjithneuro.com**). It traces the project from the initial code commit through continuous integration workflows, domain mapping, and advanced performance tuning.

## 1. Infrastructure & Hosting Setup

### Modern CI/CD Architecture

The website utilizes a modern, automated deployment pipeline leveraging **GitHub Actions** and **GitHub Pages**.

- **Version Control:** All source code is securely maintained in a central GitHub repository.
- **Automated Deployments:** A custom GitHub Actions workflow (`pages build and deployment`) was established. Every approved change pushed to the main branch automatically builds and deploys to the live server within seconds.
- **Custom Domain & DNS:** The site was successfully mapped to `dranjithneuro.com`. Multiple DNS propagation checks and CNAME re-configurations were performed to ensure stable apex domain resolution.
- **SSL/TLS Security:** The live environment is fully secured with an SSL certificate, enforcing HTTPS encryption for all visitor traffic.

## 2. Development & Optimization Timeline (Git History)

Below is the chronological sequence of major development milestones based on the repository's commit history.

### Phase 1: Foundation & Infrastructure (May 14)

#### Initial Codebase & CI/CD Setup

The core HTML, CSS, and asset files were committed to the repository `66879e2`. Immediately following, the GitHub Actions workflow was introduced `c325b7d` to enable seamless static hosting deployment.

### Phase 2: Domain Configuration (May 16 - May 17)

#### CNAME & DNS Stability Testing

A rigorous series of CNAME record creations and adjustments `08c1366` to `4ed0dbd` were executed to stabilize the custom domain connection (drranjithneuro.com) with the GitHub Pages IP addresses.

### Phase 3: Search Engine Optimization (May 17)

#### Comprehensive SEO Implementation

A massive update was deployed `3cb6be4` across 36 files to integrate structural SEO. This included dynamic Open Graph tags, Twitter cards, local geo-tags for Hyderabad (Manikonda, Puppalguda, Shaikpet), and `.htaccess` normalization rules.

### Phase 4: PageSpeed & Mobile Polish (May 17)

#### Performance & Accessibility Fixes

A sequence of rapid, targeted updates `ec0fdc8` to `5cd716e` were deployed to resolve mobile view challenges. This involved inlining critical CSS, asynchronously loading non-essential scripts, optimizing heavy images, and ensuring mobile layout stability.

### 3. Final State & Delivery Verification

**Project Status: Fully Operational**

|                   |                                                                     |
|-------------------|---------------------------------------------------------------------|
| Live URL:         | <a href="https://drranjithneuro.com">https://drranjithneuro.com</a> |
| Hosting Provider: | GitHub Pages (Enterprise CDN)                                       |
| Security:         | Enforced HTTPS / SSL Active                                         |
| Mobile Readiness: | Fully Responsive & Optimized                                        |
| SEO Status:       | Indexable, Localized for Hyderabad                                  |

**Ongoing Operations:** With the GitHub Actions CI/CD pipeline in place, any future updates requested by the clinic (such as new blog posts, service pages, or layout tweaks) can be pushed to the repository and will automatically go live without manual FTP uploads or server downtime.